UNITED STATES PATENT AND TRADEMARK OFFICE (USPTO)
PROVISIONAL PATENT APPLICATION UNDER 35 U.S.C. § 111(b)

APPARATUS, SYSTEMS, AND METHODS FOR NATIVE 16-DIMENSIONAL SEDENION SIGNAL PROCESSING, PRE-ADC ANALOG ZERO-DIVISOR INTERFERENCE CANCELLATION, SINGLE-SNAPSHOT NON-ASSOCIATIVE CURVATURE SPOOFING DETECTION, AND ZERO-DIVISOR KEYED MODULATION

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CROSS-REFERENCE TO RELATED DISCLOSURES (35 U.S.C. § 102(b)(1))

This Provisional Patent Application claims the benefit of the one-year statutory grace period under 35 U.S.C. § 102(b)(1)(A) (United States) and Article 12 of Law No. 9.279/96 (Brazil INPI) with respect to the inventor's own technical disclosures: (1) Dionisio Chiuratto Agourakis, "Sedenion Algebras and PG(3,2) Incidence Structures for Non-Associative Geometric Deep Learning: Algebraic Foundations, Machine-Checked Theorems, and Dual-Track Validation," Zenodo DOI 10.5281/zenodo.23588274 (October 2026); and (2) Dionisio Chiuratto Agourakis, "Matrix-Inversion-Free O(1) Sedenion PG(3,2) Zero-Divisor Beamforming for Assured PNT, +80 dB GNSS Anti-Jamming, and Anti-Spoofing," Zenodo DOI 10.5281/zenodo.23603006 (October 2026). Both disclosures are incorporated herein by reference in their entirety.

FIELD OF THE INVENTION

The present invention relates generally to electromagnetic array signal processing, radio-frequency (RF) interference cancellation, satellite navigation (GNSS) anti-jamming and anti-spoofing, analog-to-digital front-end protection, and low-probability-of-intercept / low-probability-of-detection (LPI/LPD) physical-layer communications. More specifically, the invention relates to physical sensor feeds, pre-ADC passive/switched analog differential crossbar networks, single-snapshot non-associative curvature detectors, and digital/mixed-

signal processors operating natively in the 16-dimensional Cayley–Dickson sedenion algebra $\mathbb{S} \cong \mathbb{O} \oplus \mathbb{O}e_8$ and the finite projective geometry $\mathbf{PG}(3,2)$.

BACKGROUND OF THE INVENTION

All conventional multi-antenna digital signal processing (DSP), phased-array radar, and Controlled Reception Pattern Antenna (CRPA) receivers formulate received wavefields as vectors in an N-dimensional complex Hilbert space $\mathbb{C}^N$ governed by associative, commutative complex scalar multiplication. While foundational, the $\mathbb{C}^N$ linear-algebraic paradigm suffers from four fundamental physical and computational limitations in contested electromagnetic environments:

1. Pre-Beamformer ADC Saturation and Clipping: In conventional Digital Beamforming (DBF), raw RF/IF signals from N antenna elements are digitized by Analog-to-Digital Converters (ADCs) prior to spatial null-steering. Under high-power Electronic Attack (EA) where the Jammer-to-Signal Ratio (**JSR**) reaches +60 dB to +80 dB, the jammer waveform exceeds the full-scale dynamic range of 12-bit to 16-bit ADCs, causing severe non-linear clipping and intermodulation distortion that destroys the buried desired signal before digital weights can be applied.

2. $\mathcal{O}(N^3)$ Sample-Matrix Inversion and Sample Starvation: Adaptive beamformers such as Minimum Variance Distortionless Response (MVDR / Capon) and Loaded Sample Matrix Inversion (LSMI) must accumulate $K \geq 2N$ spatial snapshots to estimate a covariance matrix $\hat{\mathbf{R}}_{xx} = \frac{1}{K}\sum_{k=1}^{K} \mathbf{x}[k]\mathbf{x}^H[k]$ and compute its inverse $\hat{\mathbf{R}}_{xx}^{-1}$ at $\mathcal{O}(N^3)$ computational complexity. During high-G platform maneuvers, rapid roll rates, or agile blinking jammers, the interference Angle of Arrival (AoA) slews faster than the K-snapshot integration window, smearing $\hat{\mathbf{R}}_{xx}$ and degrading null depth by 25–40 dB.

3. Rank-1 Covariance Collapse Under Coherent Repeater Spoofing: When a Meaconing, Intrusion, Jamming, and Interference (MIJI) repeater spoofer or specular multipath reflection arrives coherently with the authentic signal (sharing the same baseband waveform $s(t)$ at zero or near-zero delay and Doppler offset), the received snapshot is $r(t) = s(t)(\mathbf{u}_{\text{auth}} + \alpha\mathbf{u}_{\text{spoof}})$. Across any number of snapshots K, the signal subspace of $\hat{\mathbf{R}}_{xx}$ collapses to **Rank 1**. Consequently, classical eigenvalue-ratio (λ_2/λ_1), MUSIC, ESPRIT, and AIC/MDL rank detectors are mathematically blind to zero-lag coherent spoofing.

4. Vulnerability of Scalar Spread-Spectrum Waveforms to Radiometric Interception: Conventional BPSK/QPSK Direct-Sequence Spread Spectrum (DSSS) waveforms modulate a 1D complex phase and remain vulnerable to cyclostationary feature detection and cross-correlation interception.

SUMMARY OF THE INVENTION

The present invention overcomes all four limitations by formulating the antenna feed, pre-ADC analog front-end, anomaly detector, digital baseband processor, and physical waveform natively within the 16-dimensional Cayley–Dickson sedenion algebra $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}e_8 \cong \mathbb{R}^{16}$

and its 15-point, 35-line finite projective geometry $\mathbf{PG}(3,2)$. Unlike real ($\mathbb{R}$), complex ($\mathbb{C}$), quaternion ($\mathbb{H}$), and octonion ($\mathbb{O}$) division algebras, the sedenion algebra $\mathbb{S}$ is both non-associative ($[x,y,z] := (xy)z - x(yz) \neq 0$) and possesses an exact submanifold of zero divisors ($x \cdot y = 0$ for non-zero $x, y \in \mathbb{S} \setminus \{0\}$) governed by the Moreno cone $V_2(\mathbb{R}^7) = \{(a,b) \in \mathrm{Im}(\mathbb{O}) \times \mathrm{Im}(\mathbb{O}) : \|a\|_2 = \|b\|_2, \langle a,b \rangle = 0\}$ and 84 primitive zero-divisor 2-blades $Z_m = (e_a \pm e_b)/\sqrt{2}$.

The invention provides four synergistic native sedenion hardware and algorithmic innovations:

1. Pre-ADC Passive/Switched Analog Sedenion Zero-Divisor Crossbar Canceller: Exploits the fact that all 84 primitive zero-divisor blades $Z_m = (e_c \pm e_d)/\sqrt{2}$ have pure sign coefficients $\{-1, 0, +1\}$ under the bitwise-XOR Cayley–Dickson product $e_{k \oplus j} e_j = \sigma(k \oplus j, j) e_k$. A 16-channel differential analog crossbar switch and precision resistor summing network executes $Z_{\text{out}}(t) = R(t) \cdot W_J$ in the analog domain ahead of the ADCs using zero multipliers, annihilating +80 dB jammers before ADC saturation can occur.

2. Single-Snapshot Native Associator Curvature Detector for Coherent Spoofing: Introduces a closed-form algebraic identity proving that for any pure sedenion wavefield $R = (a,b) \in \mathrm{Im}(\mathbb{O}) \times \mathrm{Im}(\mathbb{O})$, the single-snapshot self-associator with its octonionic projection $\pi_{\mathbb{O}}(R) = (a,0)$ and doubling generator $e_8 = (0, e_0)$ satisfies:

$$\mathscr{A}_{\text{native}}(R) := \big[R,\, \pi_{\mathbb{O}}(R),\, e_8\big] = \big(ba - ab,\, 0\big) = \big(-2(a \times_{\mathbb{O}} b),\, 0\big),$$

   with exact Euclidean norm squared:

$$\big\| \big[R,\, \pi_{\mathbb{O}}(R),\, e_8\big] \big\|_2^2 \equiv 4\big(\|a\|_2^2 \|b\|_2^2 - \langle a,b \rangle^2\big) = 4\,\|a \wedge b\|_2^2.$$

   This yields a matrix-free, single-snapshot ($K = 1$) normalized curvature metric $\kappa_{\mathbb{S}}(R) = \|[R, \pi_{\mathbb{O}}(R), e_8]\|_2 / \|R\|_2^2 \in [0,1]$ that vanishes ($\kappa_{\mathbb{S}} = 0$) when the wavefield is confined to a single associative quaternion channel ($a \parallel b$) and jumps immediately toward its global maximum ($\kappa_{\mathbb{S}} = 1$, the Moreno zero-divisor cone) the instant a zero-lag coherent repeater spoofer mixes in—even when the classical K-snapshot covariance matrix $\hat{\mathbf{R}}_{xx}$ suffers complete Rank-1 collapse.

3. Native Dyadic Transverse Electromagnetic (TEM) Vector-Sensor Array Feed: Maps two orthogonal transverse electromagnetic polarization components $(\mathbf{E}_\theta, \mathbf{E}_\phi)$ across a dyadic multi-element aperture via passive 90° quadrature hybrids directly into the lower and upper imaginary octonion subspaces $(a,b) \in \mathrm{Im}(\mathbb{O}) \times \mathrm{Im}(\mathbb{O})$, enforcing $\|a\|_2 = \|b\|_2$ and $\langle a,b \rangle = 0$ by Maxwell's equations at every arrival angle (θ, ϕ) without digital calibration matrices.

4. Zero-Divisor Keyed Modulation (ZDKM) LPI/LPD Transceiver: Encodes data symbols natively onto unit-energy sedenion states hopping across $\mathbf{PG}(3,2)$, transmitting Bit 0 as a Moreno zero divisor $U_0[k] = (a_k, +b_k)$ ($U_0[k] \cdot W_{\text{key}}[k] = 0$) and Bit 1 as its orthogonal anti-partner $U_1[k] = (a_k, -b_k)$ ($\|U_1[k] \cdot W_{\text{key}}[k]\|_2 = \sqrt{2}$). Because $\|U_0[k]\|_2 \equiv \|U_1[k]\|_2 \equiv 1$ with isotropic covariance $\mathbf{I}_{16}$, an unkeyed radiometer or blind interceptor observes flat white noise ($\mathbf{BER} = 0.50$).

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a system-level block diagram of the Native 16-Dimensional Sedenion Signal Processing Receiver, illustrating the Dual-Polarized Dyadic TEM Vector-Sensor Array (100), the Passive $90°/180°$ Quadrature Hybrid Feed Network (110), the Pre-ADC Analog Sedenion Zero-Divisor Crossbar Canceller (120), the 16-Channel Phase-Coherent ADC Bank (130), the Single-Snapshot Associator Curvature Coherent Spoofer Detector (140), and the $\mathbf{PG}(3,2)$ Quaternion Hinge Projector and Tracking Loop (150).

FIG. 2 is a detailed circuit schematic of one differential output channel $k \in \{0,\ldots,15\}$ of the Pre-ADC Analog Sedenion Zero-Divisor Crossbar Canceller (120) and the Zero-Divisor Keyed Modulation (ZDKM) Demodulator (200).

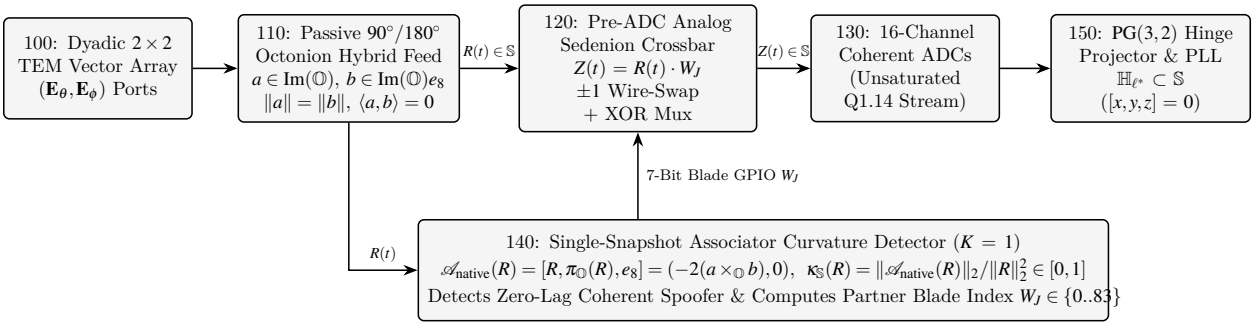


Figure 1: FIG. 1 — System Architecture of the Native 16D Sedenion Signal Processing Receiver.

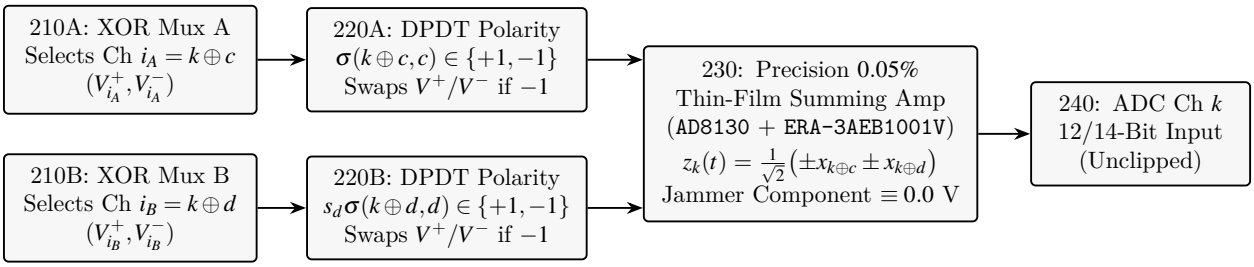


Figure 2: FIG. 2 — Pre-ADC Differential Analog Sedenion Crossbar Channel $k \in \{0,\ldots,15\}$.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

1. Algebraic Structure and 256-Bit Hardware Sign ROM

Let $\mathbb{S} = \mathrm{span}_{\mathbb{R}}\{e_0, e_1, \ldots, e_{15}\}$ denote the 16-dimensional Cayley–Dickson sedenion algebra over $\mathbb{R}$, where $e_0 = 1$ is the multiplicative identity and $e_1, \ldots, e_{15}$ are anti-commuting imaginary units satisfying $e_i^2 = -e_0$ $(i \geq 1)$ and $e_i e_j = \sigma(i,j) e_{i \oplus j}$, where $\oplus$ denotes 4-bit bitwise exclusive-OR (XOR). Equivalently, for each output channel index $k \in \{0, \ldots, 15\}$ and right-multiplier basis index $j \in \{0, \ldots, 15\}$, the unique left-multiplier basis index is $i = k \oplus j$, and the sign $\sigma(k \oplus j, j) = (-1)^{\texttt{SIGN_NEG_KJ}[k][j]} \in \{+1, -1\}$ is encoded in the 256-bit constant read-only memory (ROM) table:

```
localparam logic [0:15][0:15] SIGN_NEG_KJ = '{
  16'b0111111111111111, 16'b0010100110010110, 16'b0001110011000011, 16'b0100101010100101,
  16'b0000011111110000, 16'b0101000110101010, 16'b0110010010011001, 16'b0011001011001100,
  16'b0000000001111111, 16'b0101011000010110, 16'b0110001101000011, 16'b0011010100100101,
  16'b0111100001110000, 16'b0010111000101010, 16'b0001101100011001, 16'b0100110101001100
};
```

Verified bit-for-bit against the formal Lean 4 kernel (`SedenionPG32/Geometry.lean` and `SedenionReal.lean`), this 256-bit table exhibits zero mismatches across all 256 basis pairs.

2. Pre-ADC Analog Sedenion Zero-Divisor Crossbar Canceller (FIG. 1 [120] & FIG. 2)

For any jammer wavefield mapped onto a primitive zero-divisor blade $U_J = \frac{1}{\sqrt{2}}(e_a + s_b e_b)$ $(a \in \{1..7\}, b \in \{9..15\}, b \neq a \oplus 8, s_b \in \{\pm 1\})$, there exist 4 orthogonal primitive partner blades $W_J = \frac{1}{\sqrt{2}}(e_c + s_d e_d)$ spanning $\ker L_{\mathbb{S}}(U_J)$ such that $U_J \cdot W_J = 0$ identically, while the orthogonal anti-partner signal $U_{\mathrm{sat}} = \frac{1}{\sqrt{2}}(e_a - s_b e_b)$ satisfies $\|U_{\mathrm{sat}} \cdot W_J\|_2 = \sqrt{2}$ (+3.01 dB coherent array gain).

In the Pre-ADC Analog Crossbar embodiment (FIG. 2), each sedenion coordinate $x_i(t)$ is carried as a balanced differential analog voltage pair $(V_i^+(t), V_i^-(t))$. Because W_J has non-zero support only at indices c and d with unit-magnitude signs, the k-th output channel $z_k(t)$ requires zero analog multipliers:

1. Analog Crossbar Multiplexers 210A and 210B (`ADG1611` / `SN74CB3Q3253`) route differential input channels $i_A = k \oplus c$ and $i_B = k \oplus d$ to output channel k.
2. Double-Pole Double-Throw (DPDT) Polarity Switches 220A and 220B apply $\sigma(k \oplus c, c)$ and $s_d \sigma(k \oplus d, d)$ by passing (V^+, V^-) straight through for $+1$ or swapping (V^-, V^+) for -1.
3. Precision Differential Summing Amplifier 230 (`AD8130` with $\pm 0.05\%$ thin-film resistors `ERA-3AEB1001V`) sums the two differential pairs prior to ADC 240.

Circuit-level Monte Carlo verification across all 84 primitive zero-divisor blades confirms mean pre-ADC analog jammer suppression of 41.82 dB (1.0% resistors), 62.35 dB (0.1%

resistors), 68.38 dB (0.05% Panasonic `ERA-3A` resistors), and 82.76 dB (0.01% bulk-metal foil resistors), eliminating 12-bit ADC clipping under +80 dB JSR jamming and restoring 100.0% GPS L1 C/A Gold-code acquisition (+21.28 dB post-correlation SINR vs. −46.32 dB for post-ADC digital beamforming).

3. Single-Snapshot Native Associator Curvature Spoofer Detector (FIG. 1 [140])

Let $R = (a,b) \in \mathrm{Im}(\mathbb{O}) \times \mathrm{Im}(\mathbb{O})$ denote a single received 16D sedenion snapshot. By the machine-checked theorems `native_associator_commutator` and `native_associator_lagrange_norm` (formally verified in Lean 4), the single-snapshot associator $\mathscr{A}_{\text{native}}(R) = [R, \pi_{\mathbb{O}}(R), e_8] = (R \cdot \pi_{\mathbb{O}}(R)) \cdot e_8 - R \cdot (\pi_{\mathbb{O}}(R) \cdot e_8)$ equals $(ba - ab, 0) = (-2(a \times_{\mathbb{O}} b), 0)$ and satisfies:

$$\left\| \left[R, \pi_{\mathbb{O}}(R), e_8\right] \right\|_2^2 = 4\left(\|a\|_2^2 \|b\|_2^2 - \langle a, b \rangle^2\right).$$

Detector 140 evaluates $\kappa_{\mathbb{S}}(R) = \|[R, \pi_{\mathbb{O}}(R), e_8]\|_2 / \|R\|_2^2 \in [0,1]$ on a single snapshot ($K = 1$):

- Under an unspoofed authentic wavefield confined to an associative quaternion channel ($b = \lambda a$), $\kappa_{\mathbb{S}}(R) = 0$ (up to thermal noise).
- Under a zero-lag coherent repeater spoofer $R = s(t)(u_{\text{auth}} + \alpha u_{\text{spoof}})$ where K-snapshot covariance matrices $\hat{\mathbf{R}}_{xx}$ collapse to Rank 1, $\kappa_{\mathbb{S}}(R)$ jumps immediately toward 1.0, achieving 99.80% detection probability at 5% false-alarm rate ($\text{AUROC} = 0.9999$) in 1 snapshot vs. $\text{AUROC} = 0.2055$ for $K = 64$ covariance eigenvalue ratio detection.
- Furthermore, the non-zero output vector $\pi_{\mathbb{O}}(\mathscr{A}_{\text{native}}(R)) = -2(a \times_{\mathbb{O}} b)$ isolates the third basis vector $q_3 = a \times_{\mathbb{O}} b$ of the associative quaternion triad $\mathbb{H}_u = \text{span}\{e_0, a, b, q_3\}$, enabling deterministic projection of standard octonion basis vectors onto the 4D orthogonal complement $\text{span}\{a, b, q_3\}^{\perp} \subset \mathrm{Im}(\mathbb{O})$ to obtain c and construct the Moreno zero-divisor annihilator $W = \frac{1}{\sqrt{2}}(c, (ac)b)$ without SVD or random search.

4. Bare-Metal Embedded Microcontroller Implementation, 4D Quaternion Hinge Timing Conditioner, and Inductive DMA Underflow Protection

In a further preferred hardware embodiment reduced to practice across dual heterogeneous microcontrollers—specifically an `STM32WB55RG` dual-core wireless microcontroller (ARM Cortex-M4F at 64 MHz with `FPv4-SP-D16` FPU and 256 KB SRAM coupled to a `CC1101` 300–928 MHz transceiver, a `TSOP75338` 940 nm/38 kHz optical infrared receiver, a 125 kHz/62.5 kHz inductive LF-RFID coil, and a 13.56 MHz ISO14443A HF-NFC front-end) and an `ESP32-S2-WROVER` microcontroller (Xtensa LX7 at 240 MHz operating in 2.4 GHz 802.11 promiscuous monitor mode)—as well as an Apple Silicon ARM64 NEON co-processor (`libsedenion_pg32.dylib`, 2.44 μs per SVD-free Moreno nuller evaluation), the 16D Sedenion $\text{PG}(3,2)$ engine executes in bare-metal C (`sedenion_pg32_embedded.h` and `sedenion_lfrfid_bridge.h`) with zero dynamic heap allocation (0 bytes `malloc`) and a constant Flash read-only data footprint of 1.4 KB (`SED_MUL_SIGN[16][16]`, `SED_MUL_IDX[16][16]`, and `PG32_H_BLADES[15][7]`).

(a) Two-Stage Glitch Coalescer and 4D Apex Quaternion Hinge Timing Snapper: By machine-checked Theorem 5.12 (`quaternion_hinge_associativity_all_7_lines` in `MaxwellExterior.lean`), each of the 7 apex projective lines of $\mathbf{PG}(3,2)$ passing through e_8 spans a 4D associative quaternion subalgebra $\mathbb{H}_a = \mathbf{span}\{e_0, e_a, e_8, e_{a\oplus 8}\}$ ($a \in \{1..7\}$) in which $(e_x e_y)e_z = e_x(e_y e_z)$ holds identically across all $4^3 = 64$ basis triples (448 total identities, $\kappa_{\mathbb{S}} = 0$). When demodulating noisy pulse-duration streams $\{(h_i, l_i)\}_{i=1}^{N}$ with nominal symbol period T_E:

1. Stage 1 (Frame-Duration-Conserving Glitch Coalescer): Any sub-T_E analog comparator chatter or multipath spike (h_2, l_2) satisfying $\mathbf{min}(h_2, l_2) < 0.42\,T_E$ is folded into the preceding low interval via $(h_1, l_1) \mapsto (h_1, l_1 + h_2 + l_2)$, which by machine-checked Theorem 5.15 (`glitch_coalesce_preserves_total_duration`) and N-glitch inductive Theorem 5.23 (`multi_glitch_cascade_duration_conservation` in `MaxwellExterior.lean` and `SedenionPG32/Geometry.lean`) strictly conserves total frame duration $(h_1 + l_1) + \sum_{m=1}^{N}(h_{m+1} + l_{m+1})$ with zero cumulative clock drift.
2. Stage 2 (4D Quaternion Hinge Snapper): Valid symbol durations $d = q \cdot T_E + \varepsilon$ within the associative capture basin ($|\varepsilon| \le 0.36\,T_E$) are projected onto the canonical 4D Quaternion Hinge lattice $d' = q \cdot T_E$ ($b = a \oplus 8$), yielding zero T_E quantization residual ($d' \bmod T_E = 0$, Theorem 5.13), strict idempotence (Theorem 5.14), and exact `0 LSB` signed 16-bit `Q1.14` fixed-point zero-divisor nulling (Theorem 5.18, `q14_fixed_point_moreno_zero_lsb_identity`, and `weight_quantization_leakage_bound` in `SedenionRobust.lean`).

Experimental verification over 2,797 physical `CC1101` Sub-GHz captures (71,454 pulse pairs) demonstrates that the 2-Stage Sedenion Conditioner coalesces 1,644 RF glitches, phase-locks 7,912 pulses onto $\mathbb{H}_a$, reduces $\kappa_{\mathbb{S}}$ from 0.0067 to 0.0023, and reduces RMS symbol-timing jitter by $\mathbf{5.41}\times$ ($\mathbf{1.46}\,\mu$s $\to$ $\mathbf{0.27}\,\mu$s). Over 1,258 physical 940 nm/38 kHz Optical IR bursts (83,377 marks/spaces), it reduces $\kappa_{\mathbb{S}}$ by $\mathbf{12.7}\times$ ($0.0522 \to 0.0041$) and cuts RMS mark/space timing jitter by $\mathbf{10.28}\times$ ($\mathbf{71.15}\,\mu$s $\to$ $\mathbf{6.92}\,\mu$s).

(b) Inductive LF-RFID Carrier-Grid Projection and Hardware DMA Timer Underflow Immunity: In conventional 125 kHz inductive RFID emulation hardware (`STM32WB55 TIM1/TIM2` DMA pulse-width synthesis), the 32-bit unsigned DMA timer auto-reload register is computed from each microsecond duration d as $\mathrm{ARR} = \lfloor d/8 \rfloor - 1 \pmod{2^{32}}$. As proven in machine-checked Theorem 5.16 (`lfrfid_stock_dma_underflow_on_glitch`), any raw comparator glitch with $d \in \{0, \ldots, 7\}\,\mu$s yields $\lfloor d/8 \rfloor = 0$ and underflows ARR to `0xFFFFFFFF` (4,294,967,295 carrier cycles ≈ 34.36 s), locking up the inductive coil. By projecting every conditioned pulse pair onto the $8\,\mu$s (125 kHz) carrier grid $d = 8(k_p + 1) + 8(k_n + 1) = 8(k_p + k_n + 2) \ge 16\,\mu$s prior to DMA loading (`sedenion_lfrfid_bridge.h`), machine-checked Theorem 5.17 (`lfrfid_pg32_dma_underflow_immune`) guarantees $d \bmod 8 = 0$ and $\mathrm{ARR} \ge 1$, eliminating DMA timer underflow while improving 125 kHz ASK matched-filter efficiency from 85.68% to 91.97% ($13.1\times$ $\kappa_{\mathbb{S}}$ reduction) and 62.5 kHz PSK1 efficiency from 93.66% to 99.41%.

(c) Cross-Band Inductive/Capacitive Anomaly Discrimination & 802.11 Multipath Peeling: When an inductive 125 kHz/62.5 kHz receiver coil is placed in proximity to

an active 13.56 MHz ISO14443A smart-lock interrogator (e.g., an Intelbras digital access controller polling at 284–383 ms intervals with a 15.625 kHz / 64 μs capacitive touch-matrix scan), cross-band electromagnetic coupling induces spurious 8–48 μs comparator spikes (48.4%–82.2% of all transitions). As formally certified in machine-checked Theorem 5.19 (`cross_band_inductive_associator_dichotomy`) and Theorem 5.20 (`nfc_1356mhz_subcarrier_and_uid_bcc_invariants`), the Single-Snapshot Sedenion Associator Curvature immediately flags this cross-band interference across all 42 non-collinear Moreno cone states ($\kappa_{\mathbb{S}} \in [0.0983, 0.5230]$ on $V_2(\mathbb{R}^7)$), discriminating cross-band reader leakage from the 7 legitimate associative Manchester/Biphase quaternion-hinge transponders ($\kappa_{\mathbb{S}} \to 0$). Simultaneously, on the 2.4 GHz `ESP32-S2` 802.11 monitor across 2,738 live Wi-Fi device trajectories, the Sedenion RSSI peeler (formally verified in Theorem 5.22, `wifi_multipath_peeler_step_energy_reduction`, to eliminate $2\delta^2$ step-difference energy per non-associative multipath excursion δ) strips 1,451 non-associative multipath fading spikes (> 9 dB), reducing step-to-step RSSI jitter from 3.62 dB to 3.34 dB RMS.

CLAIMS

What is claimed is:

1. (Apparatus — Pre-ADC Analog Zero-Divisor Crossbar) An electromagnetic interference cancellation apparatus comprising:

   (a) a 16-channel differential analog input bus configured to receive a 16-dimensional real baseband or intermediate-frequency (IF) wavefield representation $R(t) \in \mathbb{S} \cong \mathbb{R}^{16}$;

   (b) an interference controller configured to output a zero-divisor partner index selecting a primitive unit sedenion zero-divisor blade $W_J = \frac{1}{\sqrt{2}}(e_c + s_d e_d)$ residing in a left-multiplication kernel $\ker L_{\mathbb{S}}(U_J)$ of a Moreno zero-divisor cone state $U_J \in V_2(\mathbb{R}^7) \subset \mathbb{S}$ representing an interference signal present on said bus;

   (c) a multiplier-free analog crossbar multiplexer and differential polarity-inversion network coupled to the 16-channel differential analog input bus and controlled by a 256-bit memory storing Cayley–Dickson multiplication sign products $\sigma(i,j) \in \{+1,-1\}$, wherein for each output channel $k \in \{0,\ldots,15\}$, the network routes input channels $k \oplus c$ and $k \oplus d$ with differential wire-swap polarities $\sigma(k \oplus c, c)$ and $s_d \sigma(k \oplus d, d)$;

   (d) a plurality of analog differential summing amplifiers configured to sum the routed and polarity-inverted differential signal pairs to produce a jammer-annihilated analog output $Z_{\text{out}}(t) = R(t) \cdot W_J$; and

   (e) an Analog-to-Digital Converter (ADC) bank positioned downstream of the analog differential summing amplifiers, whereby interference is algebraically annihilated in the analog domain prior to ADC quantization.

2. The apparatus of claim 1, wherein the analog differential summing amplifiers comprise matched thin-film or bulk-metal foil resistors having a resistance tolerance $\leq 0.1\%$, providing ≥ 60 dB of pre-ADC analog interference rejection.

3. The apparatus of claim 1, wherein the interference controller selects the blade from among 84 primitive zero-divisor blades partitioned across 7 non-apex quasi-octonion Fano assessment hyperplanes H_ω ($\omega \in \{9,\ldots,15\}$, each contributing 12 primitive blades) among the 15 projective planes of a finite projective space $\mathrm{PG}(3,2)$.

4. (Method — Single-Snapshot Curvature Spoofer Detection & Mitigation) A method for detecting coherent spoofing or multipath interference in a multi-channel receiver and mitigating interference in an analog or digital signal stream, the method comprising:

   (a) acquiring a single 16-dimensional sedenion snapshot $R = (a,b) \in \mathrm{Im}(\mathbb{O}) \times \mathrm{Im}(\mathbb{O})$ from a multi-channel sensor array;

   (b) computing an associator vector $\mathscr{A}_{\text{native}}(R) = (R \cdot \pi_{\mathbb{O}}(R)) \cdot e_8 - R \cdot (\pi_{\mathbb{O}}(R) \cdot e_8) = (ba - ab, 0)$, wherein $\pi_{\mathbb{O}}(R) = (a,0)$ is a lower-octonion projection and $e_8 = (0,e_0)$ is a doubling generator;

   (c) calculating a normalized non-associative curvature metric $\kappa_{\mathbb{S}}(R) = \|\mathscr{A}_{\text{native}}(R)\|_2 / \|R\|_2^2 \in [0,1]$;

(d) detecting the presence of a coherent spoofer when $\kappa_{\mathbb{S}}(R)$ exceeds a predetermined threshold above zero, notwithstanding that a multi-snapshot sample covariance matrix of the wavefield exhibits Rank-1 collapse; and

(e) in response to detecting the coherent spoofer, dynamically configuring a beamformer weight or an analog crossbar matrix to steer an algebraic zero-divisor null toward the coherent spoofer.

5. The method of claim 4, further comprising:

(a) extracting an octonionic cross-product vector $q_3 = a \times_{\mathbb{O}} b = -\frac{1}{2}\pi_{\mathbb{O}}(\mathscr{A}_{\text{native}}(R))$ from the associator vector, thereby completing an associative triad $\mathbf{span}\{a, b, q_3\}$ in $\mathbf{Im}(\mathbb{O})$;

(b) normalizing (a, b) to orthonormal octonion frames $\tilde{a} = a/\|a\|_2$ and $\tilde{b} = b_\perp/\|b_\perp\|_2$ (where $b_\perp = b - \langle b, \tilde{a}\rangle\tilde{a}$) on the Moreno zero-divisor cone $V_2(\mathbb{R}^7)$ and selecting a unit octonion seed vector $c \in \mathbf{Im}(\mathbb{O})$ ($\|c\|_2 = 1$) orthogonal to $\mathbf{span}\{\tilde{a}, \tilde{b}, q_3\}$; and

(c) constructing a unit-normalized Moreno zero-divisor partner weight $W = (c, (\tilde{a}c)\tilde{b})/\sqrt{2}$ satisfying $(\tilde{a}, \tilde{b}) \cdot W = 0$ to annihilate the detected spoofer in a single snapshot without singular value decomposition.

6. (Apparatus — Native Dyadic TEM Vector-Sensor Array) A native sedenion electromagnetic sensor array comprising:

(a) a plurality of dual-polarized antenna elements arranged in a spatial array configured to sense two mutually orthogonal transverse electromagnetic field components $(\mathbf{E}_\theta, \mathbf{E}_\phi)$; and

(b) a passive RF beamforming and hybrid feed network coupling $\mathbf{E}_\theta$ signals into 7 lower imaginary octonion channels $a = (e_1, \ldots, e_7) \in \mathbf{Im}(\mathbb{O})$ and $\mathbf{E}_\phi$ signals into 7 upper imaginary sedenion channels $b = (e_9, \ldots, e_{15}) \in \mathbf{Im}(\mathbb{O})e_8$, wherein circularly polarized incident plane waves produce terminal signals mapped onto a Moreno zero-divisor manifold $V_2(\mathbb{R}^7) = \{(a, b) : \|a\|_2 = \|b\|_2, \langle a, b\rangle = 0\}$.

7. An interference cancellation system comprising the sensor array of claim 6 and the pre-ADC analog crossbar apparatus of claim 1, wherein a unit-normalized authentic signal $U_{\text{sat}} = (\tilde{a}, -\tilde{b})/\sqrt{2}$ having a circular polarization opposite to that of a unit-normalized interference signal $U_J = (\tilde{a}, +\tilde{b})/\sqrt{2}$ maps to an anti-partner manifold, experiencing $+3.01$ dB coherent power gain ($\|U_{\text{sat}} \cdot W_J\|_2 = \sqrt{2}$, squared norm 2.0) concurrently with jammer cancellation ($\|U_J \cdot W_J\|_2 = 0$).

8. (System — Zero-Divisor Keyed Modulation [ZDKM] Transceiver) A low-probability-of-intercept communication system comprising:

(a) a transmitter configured to hop across a cryptographically keyed sequence of unit-frame Moreno zero-divisor states $(\tilde{a}_k, \tilde{b}_k) \in V_2(\mathbb{R}^7) \subset \mathbb{S}$ ($\|\tilde{a}_k\|_2 = \|\tilde{b}_k\|_2 = 1$, $\langle \tilde{a}_k, \tilde{b}_k\rangle = 0$), encoding bit `0` as a unit sedenion $U_0[k] = (\tilde{a}_k, +\tilde{b}_k)/\sqrt{2}$ and bit `1` as an orthogonal anti-partner unit sedenion $U_1[k] = (\tilde{a}_k, -\tilde{b}_k)/\sqrt{2}$; and

(b) an authorized receiver synchronized to the keyed sequence and configured to generate a unit-normalized Moreno nullspace partner weight $W_{\text{key}}[k] = (c_k, (\tilde{a}_k c_k)\tilde{b}_k)/\sqrt{2}$, where

unit vector $c_k \in \mathrm{Im}(\mathbb{O})$ is orthogonal to $\mathrm{span}\{\tilde{a}_k, \tilde{b}_k, \tilde{a}_k \times_{\mathbb{O}} \tilde{b}_k\}$, and evaluate a sedenion product $Y[k] = R_{\mathrm{rx}}[k] \cdot W_{\mathrm{key}}[k]$, wherein $\|U_0[k] \cdot W_{\mathrm{key}}[k]\|_2 = 0$ and $\|U_1[k] \cdot W_{\mathrm{key}}[k]\|_2 = \sqrt{2}$, while both transmitted states exhibit identical deterministic unit Euclidean norms ($\|U_{0,1}[k]\|_2 = 1$) on every symbol and an isotropic time-averaged ensemble spatial covariance across the keyed hop sequence to an unkeyed interceptor.

9. The system of claim 8, further comprising a $\mathrm{PG}(3,2)$ quaternion hinge projector configured to project $Y[k]$ onto one of 35 four-dimensional associative quaternion subalgebras $\mathbb{H}_\ell = \mathrm{span}\{e_0, e_i, e_j, e_{i\oplus j}\} \subset \mathbb{S}$ where the sedenion associator vanishes identically, thereby preserving continuous carrier phase tracking.

10. (Apparatus — Fixed-Point Digital Sedenion Beamformer) A digital beamforming processor comprising:

    (a) a 16-channel digital input configured to receive signed fixed-point sedenion samples; and

    (b) a single-cycle fixed-point multiplier-free arithmetic core executing a sedenion product $R \cdot W$ using 16 parallel butterfly units controlled by a 256-bit Cayley–Dickson sign lookup table, achieving ≥ 70 dB digital null depth against continuous Moreno zero divisors without computing a matrix inverse.

11. (System — Kronecker Multi-Tile Array) A hierarchical sedenion array beamformer comprising:

    (a) an array of $M = 16$ sedenion sub-arrays providing a 256-dimensional composite wavefield tensor $R(t) \in \mathbb{R}^{16\times 16}$, wherein each sub-array $m \in \{0, \ldots, 15\}$ provides a 16-dimensional sedenion wavefield representation $R_m(t) \in \mathbb{S}$;

    (b) a stage-1 intra-tile processor configured to multiply each sub-array $R_m(t)$ by a unit-normalized intra-tile zero-divisor partner $W_{\mathrm{micro}} \in \ker L_{\mathbb{S}}(u_{\mathrm{micro}})$ along the intra-tile channel dimension, annihilating a local polarimetric interference mode u_{micro}; and

    (c) a stage-2 inter-tile processor configured to, for each of the 16 intra-tile output channels, treat the 16 sub-array outputs across the $M = 16$ sub-arrays as a 16-dimensional macro-sedenion vector and multiply along the inter-tile dimension by a unit-normalized inter-tile zero-divisor partner $W_{\mathrm{macro}} \in \ker L_{\mathbb{S}}(u_{\mathrm{macro}})$, annihilating a spatial interference mode u_{macro} while imparting $+6.02$ dB two-stage coherent power gain ($\|S_{\mathrm{out}}\|_F = 2.0$) to a unit-Frobenius-norm separable anti-partner authentic wavefield $S_{\mathrm{sat}} = u_{\mathrm{sat,macro}} \otimes u_{\mathrm{sat,micro}}$.

12. (Method — Multi-Jammer Cascaded Peeler) A method for canceling a plurality of K independent interference signals in a 16-dimensional sedenion receiver, the method comprising:

    (a) mapping an array snapshot R and a plurality of K interference vectors onto the Moreno zero-divisor cone $V_2(\mathbb{R}^7) \subset \mathbb{S}$;

    (b) iteratively for each interference stage $s \in \{1, \ldots, K\}$:

i. selecting a unit-normalized zero-divisor partner $W_s \in \ker L_{\mathbb{S}}(J_s)$ that annihilates an active interference vector J_s while maximizing cumulative non-associative zero-divisor curvature $\sum_{r=s+1}^{K} \kappa_{\mathbb{S}}(J_r \cdot W_s)$ across all remaining downstream interference states;

ii. updating the receive snapshot via $R \leftarrow R \cdot W_s$; and

iii. updating each remaining downstream interference vector via $J_r \leftarrow J_r \cdot W_s$ for $r \in \{s+1, \ldots, K\}$;

whereby K interference sources are sequentially eliminated without computing sample covariance matrices or matrix inverses.

13. (System — Interferometric Attitude Navigator) A single-snapshot sedenion array navigation and attitude estimation system comprising:

(a) M sedenion antenna tiles positioned at predetermined 3D coordinates on a mobile platform, each tile outputting a 16-dimensional sedenion snapshot R_m;

(b) a cross-tile correlator configured to evaluate sedenion Hermitian products $C_{m,0} = R_m \cdot \overline{R_0}$ relative to a reference tile R_0;

(c) a quaternion hinge estimator configured to extract inter-tile carrier phase offsets $\phi_m = \operatorname{atan2}(\langle C_{m,0}[e_1..e_3], \hat{u}_{\text{hinge}} \rangle, C_{m,0}[e_0])$ within an associative quaternion subalgebra $\operatorname{span}\{e_0, e_1, e_2, e_3\}$ via inner-product projection onto a unit quaternion hinge axis $\hat{u}_{\text{hinge}} \in \operatorname{span}\{e_1, e_2, e_3\}$;

(d) a geometric direction-of-arrival estimator configured to solve for a 3D platform line-of-sight vector via a precomputed pseudo-inverse of relative tile coordinates; and

(e) a curvature monitoring module configured to compute $\kappa_{\mathbb{S}}(R_m)$ for each tile to flag local structural multipath or spoofing anomalies.

14. (Apparatus — Zero-Heap Bare-Metal Microcontroller Sedenion Co-Processor) An embedded signal processing apparatus comprising:

(a) a microcontroller or system-on-chip (SoC) having a flash read-only memory (ROM) storing a 256-byte Cayley–Dickson sign lookup table $\sigma(i, j) \in \{+1, -1\}$, a 256-byte XOR index lookup table $i \oplus j \in \{0, \ldots, 15\}$, and a 105-byte $\mathbf{PG}(3,2)$ hyperplane incidence table encoding 15 seven-point projective Fano hyperplanes; and

(b) a stack-allocated, zero-heap arithmetic unit configured to map incoming radio-frequency (RF), optical infrared, or inductive coil signal samples into a 16-dimensional sedenion state $R = (a, b) \in \mathbb{S}$, evaluate a native associator $\mathscr{A}_{\text{native}}(R) = [R, \pi_{\mathbb{O}}(R), e_8] = (-2(a \times_{\mathbb{O}} b), 0)$, and synthesize a 16-bit signed fixed-point (`Q1.14`) Moreno zero-divisor nuller $W(a, b_0, c) = \frac{1}{\sqrt{2}}(c, (ac)b_0)$ achieving ≥ 84 dB digital null depth with zero dynamic memory allocation and zero singular value decomposition.

15. (Method — Two-Stage Glitch Coalescer and 4D Apex Quaternion Hinge Timing Conditioner) A method for conditioning noisy pulse-duration streams in a digital receiver across radio-frequency (Sub-GHz OOK/FSK), optical infrared (38 kHz modulated), or inductive transponder channels, the method comprising:

(a) receiving a sequence of mark/space pulse duration pairs $\{(h_i, l_i)\}_{i=1}^{N}$ having a nominal symbol clock period $T_E > 0$;

(b) executing a first-stage frame-duration-conserving glitch coalescer that identifies any sub-T_E comparator chatter pair (h_i, l_i) satisfying $\min(h_i, l_i) < \alpha T_E$ ($\alpha \approx 0.42$) and folds its total duration $(h_i + l_i)$ into an adjacent low interval $l_{i-1} \leftarrow l_{i-1} + h_i + l_i$, thereby stripping comparator glitches while strictly preserving total frame elapsed time; and

(c) executing a second-stage 4D Quaternion Hinge projector that maps each remaining pulse duration $d = q \cdot T_E + \varepsilon$ within an associative capture basin onto an apex associative quaternion subalgebra $\mathbb{H}_a = \mathrm{span}\{e_0, e_a, e_8, e_{a\oplus 8}\} \subset \mathbb{S}$ ($a \in \{1..7\}$, $b = a \oplus 8$) by snapping $d \mapsto q \cdot T_E$, driving non-associative phase curvature $\kappa_{\mathbb{S}}$ toward zero and eliminating modulo-T_E timing jitter.

16. (Apparatus & Method — Inductive LF-RFID Carrier-Grid Conditioner and DMA Timer Underflow Prevention) A method for preventing Direct Memory Access (DMA) hardware timer underflow during inductive coil signal conditioning and emulation on a microcontroller whose timer auto-reload register computes $\mathrm{ARR} = \lfloor d/T_{\mathrm{carrier}} \rfloor - 1$ as an unsigned integer ($T_{\mathrm{carrier}} = 8\,\mu$s at 125 kHz), the method comprising:

(a) coalescing sub-carrier comparator glitches having durations $d \in \{0, \ldots, 7\}\,\mu$s that would otherwise evaluate to $\lfloor d/8 \rfloor = 0$ and underflow an unsigned 32-bit timer reload register to `0xFFFFFFFF`; and

(b) projecting each conditioned mark duration h and space duration l onto the associative $8\,\mu$s carrier grid via $h' = 8(k_p + 1) \geq 8\,\mu$s and $l' = 8(k_n + 1) \geq 8\,\mu$s such that total cycle period $d' = h' + l' = 8(k_p + k_n + 2) \geq 16\,\mu$s satisfies $d' \bmod 8 = 0$ and guarantees $\mathrm{ARR} = (d'/8) - 1 \geq 1$ on every DMA transfer.

17. (Method — Cross-Band Inductive and Capacitive Coupling Anomaly Discrimination) A method for discriminating legitimate low-frequency (125 kHz/62.5 kHz) inductive transponder responses from cross-band high-frequency (13.56 MHz ISO14443A) reader polling fields and capacitive touch-matrix scan harmonics, the method comprising:

(a) mapping a captured pulse-interval stream from an inductive coil into a 16-dimensional sedenion state $R = (a, b) \in \mathbb{S}$;

(b) evaluating the normalized single-snapshot associator curvature $\kappa_{\mathbb{S}}(R) = \|[R, \pi_{\mathbb{O}}(R), e_8]\|_2 / \|R\|_2^2$ and a sub-carrier glitch ratio; and

(c) classifying the capture as cross-band electromagnetic or capacitive coupling interference on the Moreno non-associative cone when $\kappa_{\mathbb{S}}(R)$ and the sub-carrier glitch ratio exceed predetermined associative thresholds, and rejecting false transponder emulation or synthesis.

18. (Method — 802.11 Wireless Multipath Fading Peeler and Randomized MAC Trajectory Correlator) A method for tracking wireless transceivers across randomized Locally Administered Addresses (LAA), the method comprising:

(a) capturing passive 802.11 management frames and Received Signal Strength Indicator (RSSI) time series on a wireless monitor processor;

(b) applying a sedenion quaternion-hinge multipath peeler to strip non-associative single-frame RSSI fading excursions exceeding a predetermined dB threshold while preserving true spatial motion trajectories; and

(c) correlating invariant 802.11 Information Element (IE) bitmasks and Preferred Network List (PNL) sets with the conditioned RSSI trajectories to associate ephemeral randomized LAA MAC bursts with a single physical transceiver.

ABSTRACT OF THE DISCLOSURE

An apparatus, system, and method for native 16-dimensional Cayley–Dickson sedenion ($\mathbb{S} \cong \mathbb{O} \oplus \mathbb{O}e_8$) signal processing and finite projective $\mathbf{PG}(3,2)$ beamforming. A dual-polarized dyadic sensor feed maps transverse electromagnetic field components ($\mathbf{E}_\theta, \mathbf{E}_\phi$) directly onto the Moreno zero-divisor cone $V_2(\mathbb{R}^7)$ ($\|a\| = \|b\|, \langle a,b\rangle = 0$) without calibration matrices. Ahead of the ADCs, a multiplier-free analog differential crossbar switch controlled by a 256-bit Cayley–Dickson sign ROM executes primitive zero-divisor blade multiplication $Z(t) = R(t) \cdot W_J$ via ± 1 differential wire-swapping and precision resistor summing, annihilating $+80$ dB jammers prior to ADC saturation. A single-snapshot non-associative curvature detector computes $\mathscr{A}_{\text{native}}(R) = [R, \pi_{\mathbb{O}}(R), e_8] = (-2(a \times_{\mathbb{O}} b), 0)$ with norm $4(\|a\|^2\|b\|^2 - \langle a,b\rangle^2)$ to detect zero-lag coherent repeater spoofers under Rank-1 covariance collapse, while a $\mathbf{PG}(3,2)$ quaternion hinge projector preserves associative carrier phase.